

Yael Konforti

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Hebrew ◇ English ◇ Spanish

RESEARCH PROFILE

PhD researcher at the University of Cambridge investigating algorithmic bias detection and mitigation through interpretability — from concept-based error slice discovery (ICML 2026) towards SAE-based pruning of spurious representations. This extends an earlier interpretability arc from an MSc at Ben-Gurion University, which modelled hidden-layer activity as visual words to trace each class's or input's major internal inference path (ECCV 2020; IEEE TPAMI 2022). Research interests: model internals, mechanistic and concept-based interpretability, failure-mode discovery and probabilistic model interpretation.

EDUCATION

Ph.D. in Computer Science and Technology

University of Cambridge, UK

Oct 2024 – Present

Thesis focus: *Algorithmic Bias Mitigation using XAI methods for AI Applications.*

Supervisors: Prof. Mateja Jamnik & Dr. Mateo Espinosa Zarlenga.

MSc in Industrial Engineering and Management (Applied Statistics Spec.)– Distinction

Ben-Gurion University, Israel | “Meitar” excellence fast track

Oct 2017 – Oct 2019

Thesis: *Probabilistic Interpretation and Visualization of Deep Neural Networks.*

Supervisors: Dr. Aharon Bar Hillel & Prof. Boaz Lerner.

GPA: 96/100

BSc in Industrial Engineering and Management– Magna Cum Laude (Top 5%)

Ben-Gurion University, Israel

Oct 2014 – Aug 2018

GPA: 88/100

PUBLICATIONS

- **Konforti, Y.**, et. al. (2026) CB-SLICE: Concept-Based Interpretable Error Slice Discovery. Proceedings of the 43rd International Conference on Machine Learning (ICML 2026), to appear.
Uses Concept Bottleneck Models to detect and characterise the model failure modes.
- **Konforti, Y.**, et. al. (2022). SIGN: Statistical Inference Graphs based on probabilistic Network activity interpretation. IEEE Transactions on Pattern Analysis and Machine Intelligence.
Models the layers' activities as probabilistic interpretable states using Hidden Markov Model and estimates inter-layer transition structure to trace inference paths through the network.
- **Konforti, Y.**, et. al. (2020). Inference graphs for CNN interpretation. In European Conference on Computer Vision (pp. 69-84). Springer, Cham.
Constructs explanatory graphs of a network's internal decision path using maximum-likelihood inference over hidden-layer activations.

INDUSTRY EXPERIENCE

Research Scientist - SightX AI

Oct 2020 – Present

Real-time object detection systems, acquired by RAFAEL Advanced Defense Systems | Tel-Aviv, Israel

- Designing and deploying deep learning models optimised for real-time small object detection and tracking on infrared imagery, with the focus on edge device optimisation and efficiency.

- Managing the research group's codebase, introducing protocols for feature integration, pull-requests and version control — improving team-wide reproducibility.
- Designing the company's evaluation framework and test-set protocol, enabling rigorous, standardised model benchmarking across deployment scenarios.

Machine Learning Researcher - BioMedSU

May 2020 – Oct 2020

Tel-Aviv, Israel

- Conducting R&D on VAEs to predict the impact of drug composition on RNA sequencing data.

TEACHING & SUPERVISION EXPERIENCE

MPhil Student Supervision (Co-supervisor)

2025 – Present

Department of Computer Science, University of Cambridge

- Mentoring an MPhil dissertation on mitigating spurious correlations in AI models; guiding research methodology, experimental design, and academic writing.

Teaching Assistant, Explainable AI (Part III/M.Phil)

Jan 2025 – Present

Department of Computer Science, University of Cambridge

- Conducting lab demonstrations, guiding students in practical applications of XAI.
- Assessing coursework and assignments.

Teaching Assistant, Production Planning and Control 2

Mar 2017 – Oct 2019

Department of Industrial Engineering and Management, Ben-Gurion University

- Conducting weekly practices for a 40 student class.
- Developing course materials, including lectures, projects, and assessments for undergraduate and post-graduate level courses.

CONFERENCE PRESENTATIONS

The 43rd International Conference on Machine Learning, Seoul, South Korea *2026*

The Women in Machine Learning Symposium @ ICML, Seoul, South Korea *2026*

The 1st Conference of International Israel Data Science Initiatives, Haifa, Israel *2022*

The 18th Israel Conference of Computer Vision, Tel-Aviv, Israel *2020*

The 16th European Conference on Computer Vision, Glasgow, Scotland *2020*

AWARDS AND SCHOLARSHIPS

The Moses Lara Trust, *2024 – Present*

Annual life stipend funding.

Faculty Excellence Scholarship, *2017 – 2019*

IEM Department, Ben Gurion University.

Full tuition and life stipend scholarship.

Excellence Scholarship for Women in STEM Fields, *2018*

The Israeli Ministry of Science and Technology.

Outstanding Instructor Award, *2018*

IEM Department, Ben-Gurion University

ACADEMIC SERVICE

Member, Women@CL Initiative, University of Cambridge *2025 – Present*

Organiser, 11th & 12th Conference of Oxbridge Women in Computer Science,

University of Cambridge *2025, 2026*